

Matteo Sacchi

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Born July 30, 1993 – Lecco, Italy

Citizenship Italian

Research profile My research focuses on non-perturbative aspects of quantum field theory, especially in supersymmetric settings, with particular emphasis on dualities, correspondences between quantum field theories and lower-dimensional systems, generalized symmetries and anomalies, dimensional compactifications, and exact observables. I am also interested in the relations between quantum field theory, conformal field theory, geometry, and string theory.

Inspire <https://inspirehep.net/authors/1727152>

Academic Positions Simons Center for Geometry and Physics, Stony Brook University
Research Assistant Professor (postdoctoral research appointment)
September 2024 - present.

University of Oxford, Mathematical Institute
Postdoctoral Research Associate in Mathematical and Theoretical Physics
November 2021 - August 2024.

Education

Ph.D. in Theoretical Physics
Università degli Studi di Milano-Bicocca (2018-2021)
Final mark: passed *cum laude*
Thesis: “Aspects of dualities and symmetry enhancements in three and four dimensions” ([link](#))
Supervisor: Prof. Sara Pasquetti

Master of Science in Theoretical Physics
Università degli Studi di Milano-Bicocca (2015-2018)
Final mark: 110/110 *cum laude*
Thesis: “Mirror Dualities in 3d and 2d Supersymmetric Gauge Theories”
Supervisor: Prof. Sara Pasquetti

Bachelor of Science in Physics
Università degli Studi di Milano-Bicocca (2012-2015)
Final mark: 110/110 *cum laude*
Thesis: “Instantons and Quantum Mechanics”
Supervisor: Prof. Alberto Zaffaroni

Awards “Sergio Fubini” prize for the best Ph.D. thesis in Theoretical Physics achieved in Italy in the year 2022.

“Yassen Stanislavov Stanev” prize for the best master thesis in Theoretical Physics achieved in Italy in the year 2018.

Organization of seminars and conferences	Forthcoming: Organization of the “XXII Avogadro Meeting on Strings, Supergravity and Gauge Theories”, Genova, 16-18 December 2026.
	Organization of the “XXI Avogadro Meeting on Strings, Supergravity and Gauge Theories”, Catania, 17-19 December 2025.
	Organization of the String Theory Journal Club, Mathematical Institute, Oxford University, academic year 2021-2022.
Teaching and Mentoring	Exercise sessions for the course “Quantum Mechanics”, Year 3 of Bachelor Degree in Physics, Università degli Studi di Milano-Bicocca, academic years 2019-2020 and 2020-2021, 40 contact hours per year. Research mentoring of PhD students through several collaborative projects, resulting in publications and preprints. In particular, conceived and supervised a project involving two PhD students at Stony Brook University that led to the preprint <i>Mirror Symmetry on a Circle</i> (2026).
Refereeing	Physical Review Letters, Physical Review D, SciPost, Journal of High Energy Physics, Nuclear Physics B.
Preprints	P3. A. Braun, O. Lewis, M. Sacchi and S. Schäfer-Nameki, <i>G_2 Manifolds from $4d \mathcal{N} = 1$ Quivers</i>, 2608.21238. P2. S. Chen, S. H. Fadda and M. Sacchi, <i>Mirror symmetry on a circle</i>, 2606.13775. P1. L. Rastelli, B. Rayhaun, M. Sacchi and G. Zafrir, $2 + 2 = 4$, 2601.00058.
Peer-Reviewed Publications	34. O. Lewis, M. Mezei, M. Sacchi and S. Schäfer-Nameki, <i>Schur Connections: Chord Counting, Line Operators, and Indices</i>, <i>SciPost Phys.</i> 20 (2026) 160, [2506.17384]. 33. A. Ardehali, D. Gang, N. J. Rajappa and M. Sacchi, <i>3d SUSY enhancement and non-semisimple TQFTs from four dimensions</i>, <i>JHEP</i> 09 (2025) 179, [2411.00766]. 32. S. Benvenuti, R. Comi, S. Pasquetti and M. Sacchi, <i>Deconfinements, Kutasov–Schwimmer dualities and $D_p[SU(N)]$ theories</i>, <i>JHEP</i> 04 (2025) 056, [2407.11134]. 31. E. Nardoni, M. Sacchi, O. Sela, G. Zafrir and Y. Zheng, <i>Dimensionally Reducing Generalized Symmetries from $(3+1)$-Dimensions</i>, <i>JHEP</i> 07 (2024) 110, [2403.15995]. 30. S. Giacomelli, C. Hwang, F. Marino, S. Pasquetti and M. Sacchi, <i>Probing bad theories with the dualization algorithm II</i>, <i>JHEP</i> 07 (2024) 165, [2401.14456]. 29. C. Beem, M. Martone, M. Sacchi, S. Singh and J. Stedman, <i>Simplifying the Type A Argyres–Douglas Landscape</i>, <i>JHEP</i> 12 (2025) 144, [2311.12123].

28. S. Giacomelli, C. Hwang, F. Marino, S. Pasquetti and **M. Sacchi**, *Probing bad theories with the dualization algorithm I*, *JHEP* **04** (2024) 008, [[2309.05326](#)].
27. S. Bajeot, S. Benvenuti and **M. Sacchi**, *S-confining gauge theories and supersymmetry enhancements*, *JHEP* **08** (2023) 042, [[2305.10274](#)].
26. **M. Sacchi**, O. Sela and G. Zafrir, *5d to 3d compactifications and discrete anomalies*, *JHEP* **10** (2023) 185, [[2305.08185](#)].
25. A. P. Braun, E. Sabag, **M. Sacchi** and S. Schäfer-Nameki, *G_2 -Manifolds from 4d $N=1$ Theories, Part I: Domain Walls*, *SciPost Phys.* **17** (2024), no. 4, 102, [[2304.01193](#)].
24. **M. Sacchi**, O. Sela and G. Zafrir, *Trinions for the 3d compactification of the 5d rank 1 E_{N_f+1} SCFTs*, *JHEP* **06** (2023) 085, [[2301.06561](#)].
23. R. Comi, C. Hwang, F. Marino, S. Pasquetti and **M. Sacchi**, *The $SL(2, \mathbb{Z})$ dualization algorithm at work*, *JHEP* **06** (2023) 119, [[2212.10571](#)].
22. N. Mekareeya and **M. Sacchi**, *Mixed anomalies, two-groups, non-invertible symmetries, and 3d superconformal indices*, *JHEP* **01** (2023) 115, [[2210.02466](#)].
21. E. Sabag and **M. Sacchi**, *A 5d perspective on the compactifications of 6d SCFTs to 4d $\mathcal{N}=1$ SCFTs*, *JHEP* **12** (2022) 017, [[2208.03331](#)].
20. M. J. Kang, C. Lawrie, K.-H. Lee, **M. Sacchi** and J. Song, *Higgs branch, Coulomb branch, and Hall-Littlewood index*, *Phys. Rev. D* **106** (2022), no. 10 106021, [[2207.05764](#)].
19. L. E. Bottini, C. Hwang, S. Pasquetti and **M. Sacchi**, *Dualities from dualities: the sequential deconfinement technique*, *JHEP* **05** (2022) 069, [[2201.11090](#)].
18. E. Maduli, N. Mekareeya and **M. Sacchi**, *Zero-form and one-form symmetries of the ABJ and related theories*, *JHEP* **04** (2022) 126, [[2112.09531](#)].
17. **M. Sacchi**, O. Sela and G. Zafrir, *On the 3d compactifications of 5d SCFTs associated with $SU(N+1)$ gauge theories*, *JHEP* **05** (2022) 053, [[2111.12745](#)].
16. C. Hwang, S. Pasquetti and **M. Sacchi**, *Rethinking mirror symmetry as a local duality on fields*, *Phys. Rev. D* **106** (2022), no. 10 105014, [[2110.11362](#)].
15. L. E. Bottini, C. Hwang, S. Pasquetti and **M. Sacchi**, *4d S-duality wall and $SL(2, \mathbb{Z})$ relations*, *JHEP* **03** (2022) 035, [[2110.08001](#)].
14. **M. Sacchi**, O. Sela and G. Zafrir, *Compactifying 5d superconformal field theories to 3d*, *JHEP* **09** (2021) 149, [[2105.01497](#)].
13. C. Hwang, S. S. Razamat, E. Sabag and **M. Sacchi**, *Rank Q E-string on spheres with flux*, *SciPost Phys.* **11** (2021), no. 2 044, [[2103.09149](#)].
12. S. Giacomelli, N. Mekareeya and **M. Sacchi**, *New aspects of Argyres–Douglas theories and their dimensional reduction*, *JHEP* **03** (2021) 242, [[2012.12852](#)].
11. C. Hwang, S. Pasquetti and **M. Sacchi**, *Flips, dualities and symmetry enhancements*, *JHEP* **05** (2021) 094, [[2010.10446](#)].
10. E. Maduli, N. Mekareeya and **M. Sacchi**, *Marginal operators and supersymmetry enhancement in 3d S-fold SCFTs*, *JHEP* **12** (2020) 017, [[2009.10123](#)].
9. E. Maduli, S. Giacomelli, N. Mekareeya and **M. Sacchi**, *3d mirrors of the circle reduction of twisted A_{2N} theories of class S*, *JHEP* **09** (2020) 161, [[2007.05019](#)].

8. **M. Sacchi**, *New 2d $\mathcal{N} = (0, 2)$ dualities from four dimensions*, *JHEP* **12** (2020) 009, [[2004.13672](#)].
7. I. Garozzo, N. Mekareeya, **M. Sacchi** and G. Zafrir, *Symmetry enhancement and duality walls in 5d gauge theories*, *JHEP* **06** (2020) 159, [[2003.07373](#)].
6. C. Hwang, S. Pasquetti and M. Sacchi, *4d mirror-like dualities*, *JHEP* **09** (2020) 047, [[2002.12897](#)].
5. I. Garozzo, N. Mekareeya and **M. Sacchi**, *Duality walls in the 4d $\mathcal{N} = 2$ $SU(N)$ gauge theory with $2N$ flavours*, *JHEP* **11** (2019) 053, [[1909.02832](#)].
4. S. Pasquetti, S. S. Razamat, **M. Sacchi** and G. Zafrir, *Rank Q E -string on a torus with flux*, *SciPost Phys.* **8** (2020) 014, [[1908.03278](#)].
3. I. Garozzo, G. Lo Monaco, N. Mekareeya and **M. Sacchi**, *Supersymmetric Indices of 3d S -fold SCFTs*, *JHEP* **08** (2019) 008, [[1905.07183](#)].
2. S. Pasquetti and **M. Sacchi**, *3d dualities from 2d free field correlators: recombination and rank stabilization*, *JHEP* **01** (2020) 061, [[1905.05807](#)].
1. S. Pasquetti and **M. Sacchi**, *From 3d dualities to 2d free field correlators and back*, *JHEP* **11** (2019) 081, [[1903.10817](#)].

Invited Conference Talks

2 + 2 = 4: a 2d/2d unitary/non-unitary correspondence,
 “Quivers, Symmetries and SCFTs 2026”,
 SISSA, Trieste, August 31 2026 ([link](#)).

A 2d/2d unitary/non-unitary correspondence,
 “Aspects of Supersymmetric Quantum Field Theory 2025”,
 SNU, Seoul, November 12 2025.

Anomalies, generalized symmetries, and compactifications,
 “XIV Workshop on Geometric Correspondences of Gauge Theories”,
 iGAP, Trieste, June 27 2024 ([link](#)).

Kutasov–Schwimmer-like dualities from deconfinement,
 “Geometry and Symmetries of SCFTs”,
 MITP, Mainz, June 17 2024.

SUSY QFTs in various dimensions,
 “XVIII Avogadro Meeting”,
 Torino, December 22 2022.

Rethinking mirror symmetry as a local duality on fields,
 “Chennai-Southampton Workshop”,
 Southampton, March 28 2022.

New 2d $\mathcal{N} = (0, 2)$ dualities from four dimensions,
 “GLSMs – 2020”,
 Zoom, August 18 2020 ([link](#)).

From 3d dualities to 2d free field correlators and back,
 “TFI 2019: Theories of Fundamental Interactions 2019”,
 Torino, October 21-23 2019.

Invited Seminars

2 + 2 = 4: a 2d/2d unitary/non-unitary correspondence,
 City College of New York, May 8 2026.

2 + 2 = 4: a 2d/2d unitary/non-unitary correspondence,
Perimeter Institute, February 3 2026 ([link](#)).

Chord diagrams for Schur indices,
University of Milano-Bicocca, June 27 2025.

Schur/q-oscillators correspondence: generalized chord diagrams and Toda chain,
Seminar series “Interfaces of Physics and Mathematics”,
Simons Center for Geometry and Physics, May 20 2025.

Symmetries, anomalies, and compactifications in QFT,
University of Sheffield, May 17 2024.

Anomalies, generalized symmetries, and compactifications,
Online seminar series “Quiver meetings”, April 19 2024 ([link](#)).

Aspects of dualities and symmetry enhancements in three and four dimensions,
Presidenza INFN, Rome, September 25 2023.

Discrete anomalies and generalized symmetries from 3d superconformal indices,
DESY Hamburg, June 19 2023.

Mixed anomalies and generalized symmetries from 3d superconformal indices,
SISSA Trieste, February 14 2023.

Mixed anomalies and generalized symmetries from 3d superconformal indices,
Birmingham, December 13 2022.

Mixed anomalies and generalized symmetries from 3d superconformal indices,
Neve Shalom, November 29 2022.

Mixed anomalies and generalized symmetries from 3d superconformal indices,
Imperial College London, November 16 2022.

Compactifications of 6d SCFTs to 4d and duality domain walls in 5d gauge theories,
Swansea, June 6 2022.

4d S-walls and mirror symmetry as a local duality on fields,
Oxford, November 8 2021.

New dualities from dimensional reductions and uplifts,
Online seminar series “Strings and QFTs for Eurasian time zone”,
August 26 2020 ([link](#)).

3d mirrors of the circle reduction of twisted A_{2N} theories of class S,
Online seminar series “Quiver meetings” of Imperial College London,
August 21 2020.

The $E[USp(2N)]$ theory: E-string on tori and 4d mirror-like dualities,
SISSA Trieste, February 19 2020.

Mirror Dualities in 3d and 2d Supersymmetric Gauge Theories,
Divulgative seminar at the award ceremony for the “Stanev Prize 2019”,
Roma “Tor Vergata”, November 22 2019.

Languages

Italian

Level: Native

English

Level: Professional